+46 070 013 7921
santiagodp1990@gmail.com
santimda.github.io
santiago-del-palacio
santimda
Gothenburg, Sweden

Santiago del Palacio

Data Scientist & Astrophysicist

Professional Summary

Data Scientist, Astrophysicist & Scientific Software Developer with 10+ years of experience in statistical modelling, machine learning, and Python-based data analysis applied to complex, large-scale datasets. Strong proficiency in Python and SQL, with hands-on experience in BigQuery, data pipeline design, model building and validation, uncertainty quantification, and data visualisation. Experienced in developing end-to-end analytical workflows, and building reproducible scientific software and automated data-processing pipelines. Proven ability to lead technical projects, communicate complex results clearly, and collaborate across disciplines.
Author of 8 first-author and 60+ co-authored peer-reviewed publications, including papers in Nature and Science Advances. Principal Investigator on 18 and Co-Investigator on 26+ competitively awarded international research projects. Delivered 18+ conference presentations internationally.
Currently seeking data science roles where I can apply statistical modelling, machine learning, and computational expertise to complex engineering and real-world data problems, with particular interest in predictive modelling, time-series analysis, data-driven decision-making, and building reproducible and reusable software solutions.

Work Experience

04/2022 – 03/2026 **Postdoctoral Research Fellow**, *Chalmers University of Technology, Dept. of Space, Earth and Environment*, Gothenburg, Sweden

- Led an independent, cross-disciplinary research programme, defining scientific direction and coordinating collaborations across multiple institutions. This highly competitive fellowship also required to work towards bridging multiple research groups within the division.
- Developed and validated physical models using statistical inference and numerical methods; quantified parameter uncertainties using Bayesian and frequentist approaches; developed tools for interpreting multi-wavelength observational datasets (radio, mm-wave, X-rays).
- Co-supervised 1 Ph.D. student, providing guidance in research design, data analysis, and problem-solving.
- Organised weekly departmental seminars to improve engagement and knowledge sharing within the division.
- Served as referee for major journals and as member of international time-allocation committees for ALMA and *XMM-Newton*, evaluating project quality, technical feasibility and scientific impact, and providing constructive and actionable feedback.

04/2019 – 03/2022 **Postdoctoral Research Fellow (CONICET)**, *Argentine Institute of Radioastronomy*, Buenos Aires, Argentina

- Led an independent research programme, including project design, development of theoretical and numerical models, and coordinating international collaborations and multi-site observational campaigns.
- Built automated data processing pipelines in Python for pulsar monitoring within the PuMA Collaboration.
- Supervised 3 M.Sc. students, defining project scope, work strategy, and continuous mentoring.

04/2014 – 03/2019 **Ph.D. Research Fellow (CONICET)**, *National University of La Plata*, La Plata, Argentina

- Developed numerical radiation-transport codes (FORTRAN) to model non-thermal emission in astrophysical systems.
- Designed and executed multi-observatory campaigns; analysed and interpreted data from radio, X-ray, and gamma-ray facilities.
- Awarded the *Carlos M. Varsavsky Prize* for best Ph.D. thesis in astronomy in Argentina (2018–2019 cohort).

02/2011 – 03/2022 **Teaching Assistant / Head of Practical Assignments**, *National University of La Plata*, La Plata, Argentina

Taught several courses on Calculus, Algebra, and Astrophysics at undergraduate and graduate level for both science and engineering students. Student evaluations consistently remarked my clarity at explaining complex concepts and good predisposition overall.

Education

04/2014 – 11/2018 **Ph.D. in Astronomy**, *National University of La Plata*, La Plata, Argentina, Mark: 10/10
Thesis: *Non-thermal emission from systems harbouring massive stars*.
Award: *Carlos M. Varsavsky Prize* (best Ph.D. thesis in astronomy in Argentina, 2018–2019).

03/2008 – 03/2014 **Licenciatura in Astronomy (equiv. B.Sc. + M.Sc.)**, *National University of La Plata*, La Plata, Argentina, GPA: 9.56/10
Award: *Joaquín V. González Prize* (distinction for highest-achieving students, La Plata City Council).

Technical Skills

Programming	Python (advanced: NumPy, SciPy, Astropy, scikit-learn, Matplotlib, Pandas), FORTRAN (advanced), SQL (proficient), Bash (intermediate), R (basic)
Tools & Platforms	Git/GitHub, Linux/Unix, LaTeX, Jupyter, BigQuery, Looker Studio, LaTeX; scientific data-analysis pipelines; AI-assisted development (GitHub Copilot, LLMs, OpenCode/agentic workflows)
Methods	Numerical simulations, statistical modelling, uncertainty quantification, Bayesian inference, causal inference, numerical methods, data pipeline design, data visualisation, scientific image analysis, time-series analysis, signal processing, machine learning, dashboards
Soft Skills	Project management, scientific writing, public communication, international and interdisciplinary collaboration, student supervision

Languages

Spanish	Mother tongue
English	Fluent (C2)
Swedish	Upper Intermediate (B2/C1)

Outreach & Science Communication

2015 – 2026 14 public outreach activities, including public talks on astronomy and astrophysics (e.g. *Neutron Stars and Pulsars*, Slottsskogsobservatoriet, Gothenburg, 2026), lectures at the Swedish National Space School *Rymdforskarskolan* (2025), guided visits at open observatory nights, and co-authoring press releases.

References

Colleagues and line managers have highlighted my independence, contribution to a positive work environment, and capacity to move across different fields. References available upon request.